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MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours)
REGULAR EXAMINATION October 2026
Fifth SEMESTER
Discipline Specific Core Course (DSC)
MG5DSCBTS300 - DEVELOPMENTAL BIOLOGY
(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I)and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A.MCQ Answer all questions. Each question carries1 marks

1. Who proposed the Germ Plasm Theory?

a). August Weismann [K] / [CO1]

2. During normal cleavage, the chromosome number of blastomeres is:

c). Maintained [K] / [CO2]

3. The darker pigmented region of a frog egg is called the:

c). Animal hemisphere [K] / [CO4]

4. An egg containing a large amount of yolk is called:

c). Macrolecithal [U] / [CO2]

5. The lens develops from the:

b). Surface ectoderm [K] / [CO4]

6. In epiboly, cells mainly:

b). Spread over other cells [U] / [CO2]

7. Which of the following best describes the focus of chemical developmental biology?

b). Chemical substances and molecular interactions [K] / [CO1]

8. Which process produces specialized cell types during development?

a). Cell differentiation [K] / [CO5]

9. Salamanders show greater regenerative ability than adult mammals mainlybecause they can:

c). Re-form complex tissues after injury [K] / [CO3]

10. A commonly used method for constructing fate maps involves:

a). Cell marking [U] / [CO2] [1x10 = 10]

Part B. Short Answer Questions.

Answer any 10 questions. Each question carries 2 marks

11. **Name the major phases of the menstrual cycle.** [K] / [CO2]

The major phases are:

1. Menstrual phase – Shedding of the uterine lining.

2. Follicular/proliferative phase – Growth of ovarian follicles and regeneration of the uterine lining.

3. Ovulation – Release of the mature ovum from the ovary.

4. Luteal/secretory phase – The corpus luteum secretes progesterone and prepares the uterus for implantation**. (4 major phases-1/2 mark each)**

12. **A female insect produces offspring without mating. Identify the reproductive process and explain your answer.** [A] / [CO2]

The process is parthenogenesis. It is a type of reproduction in which an offspring develops from an unfertilized egg without fertilization by a male gamete Eg-. Honeybees and Ants **(Identification-1 mark; explanation-1 mark)**

13. **What is the source of alpha-fetoprotein during pregnancy?** [K] / [CO3]

Alpha-fetoprotein (AFP) is produced mainly by the fetal liver, with an initial contribution from the yolk sac. It is present in fetal blood and can be detected in maternal blood and amniotic fluid. (source-2 marks)

14.Des**cribe neurulation and notogenesis in frog embryo.** [K] / [CO4]

Neurulation: The dorsal ectoderm forms the neural plate, which folds and closes to form the neural tube. The neural tube develops into the central nervous system.

Notogenesis: Formation of the notochord from axial mesoderm. The notochord provides support and helps organize the development of the neural tube.

(Neurulation – 1 mark; Notogenesis– 1 mark)

15. What is the apical ectodermal ridge (AER)? [K] / [CO3]

Ans: The apical ectodermal ridge is a thickened band of ectoderm present at the distal tip of a developing vertebrate limb bud. It secretes signals, including FGF, that maintain limb-bud growth and regulate development along the proximodistal axis**. (Definition-1 mark; role-1 marks)**

16. Na**me the main parts of a sperm.** [K] / [CO2]

The main parts of a sperm are:

1. Head

2. Acrosome

3. Neck

4. Middle piece

5. Tail

17. **What is the zona pellucida**? [U] / [CO2]

Ans: The zona pellucida is a transparent, non-cellular glycoprotein layer surrounding the mammalian oocyte. It helps in sperm binding, species-specific fertilization, and prevention of polyspermy after fertilization. **(Definition – 1 mark; Any one function – 1 mark)**

18. **Mention any two biochemical changes occurring during frog**

metamorphosis. [K] / [CO4]

1. Protein metabolism: Breakdown of larval proteins and synthesis of adult proteins.

2. Carbohydrate metabolism: Changes in glycogen breakdown and glucose utilization for energy.

3. Lipid metabolism: Breakdown of stored fats to meet energy requirements.

4. Enzyme activity: Increased lysosomal enzyme activity helps in the degeneration and resorption of the tadpole tail.

5. Nucleic acid metabolism: Changes in DNA and RNA synthesis regulate cell growth, differentiation and tissue remodelling.

6. Respiratory metabolism: Development of lungs and increased capacity for pulmonary respiration.

7. Nitrogen metabolism: Shift from mainly ammonotelic excretion in tadpoles towards increased ureotelic excretion in adult frogs.

8. Hormonal changes: Thyroid hormones, especially thyroxine (T4) and triiodothyronine (T3), control metamorphosis. **(Any four points-2marks)**

19. Name any two derivatives of endoderm. [K] / [CO2]

Ans: Epithelial lining of the digestive tract, Epithelial lining of the respiratory tract, Liver and Pancreas. (1 mark for each)

20. Define pregnancy. [K] / [CO2]

Pregnancy is the physiological condition in which an embryo or fetus develops within the uterus following implantation, until birth.

(Development of embryo/fetus in the uterus – 1 mark; Implantation to birth or equivalent correct description – 1 mark)

21. **Define prenatal diagnosis.** **[K] / [CO3]**

Prenatal diagnosis refers to the examination of an embryo or fetus before birth to detect genetic, chromosomal, or developmental abnormalities. Examples include ultrasonography, amniocentesis, and chorionic villus sampling.

(Definition – 1 mark; Any one suitable method or purpose – 1 mark)

22.Name the three major regions into which the mesoderm differentiates.

Ans: The three major regions are: Axial mesoderm, Paraxial mesoderm, Lateral plate mesoderm (three major region-full marks) [K] / [CO4] , [2x10 = 20]

Part C. Short Essay Questions

Answer any 5questions. Each question carries 4 marks

23. **Explain the role of segmentation genes in Drosophila development. [U] / [CO5]**

Segmentation genes regulate the formation of body segments in the developing Drosophila embryo. They act in a hierarchical sequence to establish the body plan.

1. Gap genes – Divide the embryo into broad regions along the anterior–posterior axis. Examples: hunchback and Krüppel.

2. Pair-rule genes – Establish alternating stripes of gene expression. Example: even-skipped.

3. Segment-polarity genes – Establish and maintain the anterior–posterior organization within each

segment. Example: engrailed.

(Introduction – 1 mark; 3 Points – 1 mark each)

24. **Differentiate between gametogenesis and spermatogenesis. [K] / [CO2]**

Gametogenesis	**Spermatogenesis**
General process of formation of gametes	Specific process of formation of sperm.
Includes spermatogenesis and oogenesis	Occurs in the seminiferous tubules of the testes.
Produces male or female gametes.	Produces male gametes
Includes stages of germ-cell development and maturation; details vary by sex.	Involves multiplication, growth, meiosis and spermiogenesis.

(1 mark for each difference)

25. Explain the hormonal control of male reproductive function. [K] / [CO2]

Male reproductive function is controlled mainly by the hypothalamic–pituitary–gonadal axis.

1. GnRH: The hypothalamus releases gonadotropin-releasing hormone, which stimulates the anterior pituitary.

2. LH: Luteinizing hormone stimulates Leydig cells in the testes to produce testosterone.

3. FSH: Follicle-stimulating hormone acts on Sertoli cells and supports spermatogenesis.

4. Testosterone and inhibin: Testosterone supports male reproductive organs and secondary sexual characteristics. Inhibin from Sertoli cells suppresses FSH secretion, while testosterone provides negative feedback on the hypothalamus and pituitary. (4 Points -4 marks)

26. **Differentiate between the morula and blastula. [K] / [CO2]**

Morula	**Blastula**
A solid ball of blastomeres.	It is generally a hollow ball of cells.
It lacks a blastocoel	It contains a cavity called the blastocoel.
Commonly described around the 16–32-cell stage, though this varies by species	Usually contains more cells after further cleavage.
Forms during cleavage, before blastulation	Forms from the morula by blastulation

(Any four differences-4 marks)

27.Explain the role of seminal vesicles, prostate gland and Cowper's glands in the formation of semen. [K] / [CO2]

- Ans: Seminal vesicles: Secrete fructose-rich fluid that supplies energy to sperm and contributes a major portion of semen.

- Prostate gland: Adds fluid containing citrate, enzymes and other substances that support sperm function and contribute to semen liquefaction.

- Cowper's (bulbourethral) glands: Secrete mucus-like pre-ejaculatory fluid that lubricates the urethra and helps neutralize residual acidity.

• Overall role: These secretions combine with sperm and other reproductive tract fluids to form semen, supporting sperm transport and function.

(role of seminal vesicles, prostate gland and cowpers glands-1mark each; overall role- 1 mark)

28. Describe the role of the dorsal lip of the blastopore as an organizer. [U] / [CO4]

The dorsal lip of the blastopore in an amphibian embryo acts as an organizer during gastrulation. Its organizing role was demonstrated by Hans Spemann and Hilde Mangold.

1. Induction: It induces overlying ectoderm to develop into neural tissue.

2. Axis formation: It helps establish the main body axis of the embryo.

3. Cell movements: Cells migrating through this region contribute to axial mesoderm, including the notochord.

4. Secondary axis: When transplanted to another region of a suitable host embryo, it can induce a secondary body axis, demonstrating its organizer activity. **(4 points-3 marks)**

29. Define embryo transfer technology and describe its basic steps. [U] / [CO4]

Embryo transfer technology is an assisted reproductive technique in which an embryo is transferred into the uterus of a recipient female to establish pregnancy. Basic steps are

1. Donor preparation: A suitable donor female is selected; in many protocols, superovulation is induced.

2. Fertilization: Oocytes are fertilized naturally or by an assisted method, depending on the technique.

3. Embryo collection: Developing embryos are collected and assessed.

4. Transfer: A suitable embryo is transferred into the uterus of a synchronized recipient female for implantation and further development. **(Definition-1 marks; steps-3 marks)**

30.Explain the different stages of oogenesis. [U] / [CO2]

Oogenesis is the process of formation and maturation of the female gamete in the ovary.

1. Multiplication phase: Oogonia divide by mitosis to increase their number.

2. Growth phase: Oogonia develop into primary oocytes, which grow and accumulate cytoplasm and nutrients.

3. Maturation phase I: The primary oocyte completes meiosis I to produce a secondary oocyte and the first polar body.

4. Maturation phase II: The secondary oocyte begins meiosis II and in humans, completes it only after fertilization, producing the ovum and another polar body. **(Different stages-4 marks, if a properly labelled diagram is drawn, give maximum 2 marks)**

[4x5= 20]